SoccerGMR Full-Decodable Test (2970/2986) — primary

Standard-trained checkpoints | Full-Decodable Test (2970/2986) | 1633 positive / 1337 negative | 16 original exclusions

Value above; (method - same-backbone GMR) below. G-mIoU: 0-1; others: percent / percentage points.

Backbone	Method	AUROC	mAP	mR@1	mR@5	mR+@5	N-acc τ=0.4	Rej-F1 τ=0.4	G-mIoU@1 τ=0.4	G-mIoU@3 τ=0.4	N-acc τ=0.5	Rej-F1 τ=0.5	G-mIoU@1 τ=0.5	G-mIoU@3 τ=0.5	N-acc τ=0.6	Rej-F1 τ=0.6	G-mIoU@1 τ=0.6	G-mIoU@3 τ=0.6
Moment-DETR	**GMR (baseline)**	**77.92**	**5.41**	**3.39**	**7.33**	**0.81**	**0.00**	**0.00**	**0.0589**	**0.0253**	**97.01**	**78.92**	**0.4885**	**0.4575**	**97.16**	**78.97**	**0.4891**	**0.4582**
	GMR + GEQ Frozen	79.03 (+1.11)	5.41 (0.00)	3.39 (0.00)	7.33 (0.00)	0.81 (0.00)	96.41 (+96.41)	78.74 (+78.74)	0.4858 (+0.4269)	0.4548 (+0.4295)	97.16 (+0.15)	78.97 (+0.05)	0.4891 (+0.0006)	0.4582 (+0.0007)	97.16 (0.00)	78.82 (-0.15)	0.4888 (-0.0003)	0.4581 (-0.0001)
	GMR + GEQ Full	78.62 (+0.70)	5.28 (-0.13)	3.31 (-0.08)	7.33 (0.00)	0.65 (-0.16)	91.92 (+91.92)	77.15 (+77.15)	0.4648 (+0.4059)	0.4340 (+0.4087)	96.63 (-0.37)	78.80 (-0.12)	0.4857 (-0.0028)	0.4550 (-0.0025)	97.08 (-0.07)	78.93 (-0.04)	0.4877 (-0.0014)	0.4570 (-0.0012)
	GMR + Multi	79.25 (+1.33)	6.06 (+0.65)	3.86 (+0.47)	8.63 (+1.30)	0.49 (-0.32)	83.62 (+83.62)	74.26 (+74.26)	0.4237 (+0.3648)	0.3964 (+0.3711)	97.16 (+0.15)	78.89 (-0.03)	0.4839 (-0.0046)	0.4563 (-0.0012)	97.16 (0.00)	78.89 (-0.08)	0.4839 (-0.0052)	0.4563 (-0.0019)
	GMR + Multi + GEQ Frozen	78.81 (+0.89)	6.06 (+0.65)	3.86 (+0.47)	8.63 (+1.30)	0.49 (-0.32)	88.93 (+88.93)	76.12 (+76.12)	0.4471 (+0.3882)	0.4196 (+0.3943)	97.16 (+0.15)	78.92 (0.00)	0.4839 (-0.0046)	0.4563 (-0.0012)	97.16 (0.00)	78.92 (-0.05)	0.4839 (-0.0052)	0.4563 (-0.0019)
	GMR + Multi + GEQ Full	78.97 (+1.05)	6.39 (+0.98)	3.38 (-0.01)	10.09 (+2.76)	0.77 (-0.04)	81.82 (+81.82)	73.40 (+73.40)	0.4196 (+0.3607)	0.3902 (+0.3649)	97.08 (+0.07)	78.86 (-0.06)	0.4870 (-0.0015)	0.4578 (+0.0003)	97.16 (0.00)	78.87 (-0.10)	0.4874 (-0.0017)	0.4581 (-0.0001)
EaTR	**GMR (baseline)**	**81.59**	**15.24**	**10.11**	**20.37**	**6.99**	**47.87**	**57.25**	**0.3208**	**0.2701**	**66.12**	**67.38**	**0.3961**	**0.3467**	**85.42**	**75.06**	**0.4739**	**0.4264**
	GMR + GEQ Frozen	82.57 (+0.98)	15.24 (0.00)	10.11 (0.00)	20.37 (0.00)	6.99 (0.00)	57.07 (+9.20)	63.40 (+6.15)	0.3577 (+0.0369)	0.3090 (+0.0389)	77.56 (+11.44)	73.13 (+5.75)	0.4421 (+0.0460)	0.3959 (+0.0492)	88.93 (+3.52)	76.71 (+1.65)	0.4870 (+0.0131)	0.4420 (+0.0156)
	GMR + GEQ Full	82.34 (+0.75)	15.27 (+0.03)	10.42 (+0.31)	20.13 (-0.24)	7.48 (+0.49)	55.80 (+7.93)	62.45 (+5.20)	0.3527 (+0.0319)	0.3021 (+0.0320)	80.25 (+14.14)	74.05 (+6.67)	0.4517 (+0.0556)	0.4047 (+0.0580)	91.55 (+6.13)	77.42 (+2.36)	0.4975 (+0.0236)	0.4512 (+0.0248)
	GMR + Multi	82.43 (+0.84)	16.12 (+0.88)	10.77 (+0.66)	21.85 (+1.48)	7.49 (+0.50)	45.33 (-2.54)	56.53 (-0.72)	0.3171 (-0.0037)	0.2610 (-0.0091)	62.30 (-3.81)	66.64 (-0.74)	0.3873 (-0.0088)	0.3331 (-0.0136)	75.02 (-10.40)	71.64 (-3.42)	0.4376 (-0.0363)	0.3855 (-0.0409)
	GMR + Multi + GEQ Frozen	83.46 (+1.87)	16.12 (+0.88)	10.77 (+0.66)	21.85 (+1.48)	7.49 (+0.50)	66.34 (+18.47)	69.13 (+11.88)	0.4007 (+0.0799)	0.3492 (+0.0791)	79.21 (+13.09)	74.21 (+6.83)	0.4515 (+0.0554)	0.4030 (+0.0563)	88.11 (+2.69)	76.97 (+1.91)	0.4868 (+0.0129)	0.4395 (+0.0131)
	GMR + Multi + GEQ Full	83.43 (+1.84)	16.81 (+1.57)	11.65 (+1.54)	21.79 (+1.42)	7.47 (+0.48)	67.17 (+19.30)	69.40 (+12.15)	0.4065 (+0.0857)	0.3537 (+0.0836)	80.33 (+14.21)	74.82 (+7.44)	0.4596 (+0.0635)	0.4092 (+0.0625)	89.01 (+3.59)	77.12 (+2.06)	0.4930 (+0.0191)	0.4438 (+0.0174)
QD-DETR	**GMR (baseline)**	**83.62**	**12.31**	**8.81**	**16.05**	**4.11**	**52.13**	**61.27**	**0.3322**	**0.2817**	**87.81**	**77.24**	**0.4793**	**0.4326**	**95.06**	**79.04**	**0.5076**	**0.4611**
	GMR + GEQ Frozen	83.14 (-0.48)	12.31 (0.00)	8.81 (0.00)	16.05 (0.00)	4.11 (0.00)	72.10 (+19.97)	71.46 (+10.19)	0.4158 (+0.0836)	0.3662 (+0.0845)	87.58 (-0.22)	76.86 (-0.38)	0.4767 (-0.0026)	0.4301 (-0.0025)	93.27 (-1.80)	78.33 (-0.71)	0.4999 (-0.0077)	0.4534 (-0.0077)
	GMR + GEQ Full	83.29 (-0.33)	13.50 (+1.19)	9.65 (+0.84)	17.70 (+1.65)	4.52 (+0.41)	77.49 (+25.36)	73.92 (+12.65)	0.4404 (+0.1082)	0.3900 (+0.1083)	90.88 (+3.07)	77.76 (+0.52)	0.4923 (+0.0130)	0.4448 (+0.0122)	94.84 (-0.22)	78.86 (-0.18)	0.5080 (+0.0004)	0.4607 (-0.0004)
	GMR + Multi	82.03 (-1.59)	12.04 (-0.27)	8.43 (-0.38)	15.54 (-0.51)	3.24 (-0.87)	54.53 (+2.39)	61.65 (+0.38)	0.3327 (+0.0005)	0.2857 (+0.0040)	87.28 (-0.52)	76.70 (-0.54)	0.4708 (-0.0085)	0.4267 (-0.0059)	93.12 (-1.94)	78.20 (-0.84)	0.4926 (-0.0150)	0.4496 (-0.0115)
	GMR + Multi + GEQ Frozen	82.44 (-1.18)	12.04 (-0.27)	8.43 (-0.38)	15.54 (-0.51)	3.24 (-0.87)	68.59 (+16.45)	69.18 (+7.91)	0.3903 (+0.0581)	0.3460 (+0.0643)	88.48 (+0.67)	77.14 (-0.10)	0.4735 (-0.0058)	0.4303 (-0.0023)	93.87 (-1.20)	78.19 (-0.85)	0.4934 (-0.0142)	0.4518 (-0.0093)
	GMR + Multi + GEQ Full	82.42 (-1.20)	11.85 (-0.46)	8.34 (-0.47)	15.26 (-0.79)	3.41 (-0.70)	88.71 (+36.57)	77.24 (+15.97)	0.4765 (+0.1443)	0.4334 (+0.1517)	95.44 (+7.63)	78.81 (+1.57)	0.5026 (+0.0233)	0.4606 (+0.0280)	96.86 (+1.80)	78.89 (-0.15)	0.5075 (-0.0001)	0.4660 (+0.0049)
CG-DETR	**GMR (baseline)**	**82.34**	**13.58**	**9.93**	**17.97**	**5.59**	**94.39**	**78.43**	**0.5068**	**0.4586**	**96.56**	**78.62**	**0.5143**	**0.4660**	**96.78**	**78.52**	**0.5150**	**0.4667**
	GMR + GEQ Frozen	82.90 (+0.56)	13.58 (0.00)	9.93 (0.00)	17.97 (0.00)	5.59 (0.00)	76.74 (-17.65)	73.29 (-5.14)	0.4360 (-0.0708)	0.3869 (-0.0717)	91.92 (-4.64)	78.28 (-0.34)	0.4982 (-0.0161)	0.4496 (-0.0164)	96.19 (-0.60)	79.19 (+0.67)	0.5141 (-0.0009)	0.4658 (-0.0009)
	GMR + GEQ Full	81.82 (-0.52)	14.01 (+0.43)	10.23 (+0.30)	18.77 (+0.80)	5.31 (-0.28)	75.39 (-19.00)	72.03 (-6.40)	0.4316 (-0.0752)	0.3797 (-0.0789)	94.32 (-2.24)	78.94 (+0.32)	0.5092 (-0.0051)	0.4593 (-0.0067)	96.86 (+0.07)	79.11 (+0.59)	0.5176 (+0.0026)	0.4690 (+0.0023)
	GMR + Multi	81.02 (-1.32)	15.48 (+1.90)	10.52 (+0.59)	20.42 (+2.45)	6.12 (+0.53)	51.16 (-43.23)	59.50 (-18.93)	0.3398 (-0.1670)	0.2823 (-0.1763)	76.66 (-19.90)	71.45 (-7.17)	0.4423 (-0.0720)	0.3887 (-0.0773)	88.86 (-7.93)	74.76 (-3.76)	0.4893 (-0.0257)	0.4378 (-0.0289)
	GMR + Multi + GEQ Frozen	83.01 (+0.67)	15.48 (+1.90)	10.52 (+0.59)	20.42 (+2.45)	6.12 (+0.53)	71.28 (-23.11)	70.59 (-7.84)	0.4200 (-0.0868)	0.3669 (-0.0917)	84.29 (-12.27)	75.79 (-2.83)	0.4727 (-0.0416)	0.4222 (-0.0438)	91.32 (-5.46)	77.33 (-1.19)	0.4986 (-0.0164)	0.4497 (-0.0170)
	GMR + Multi + GEQ Full	82.81 (+0.47)	15.93 (+2.35)	10.75 (+0.82)	21.44 (+3.47)	7.62 (+2.03)	56.10 (-38.29)	63.34 (-15.09)	0.3593 (-0.1475)	0.3041 (-0.1545)	77.94 (-18.62)	73.56 (-5.06)	0.4455 (-0.0688)	0.3947 (-0.0713)	89.75 (-7.03)	77.20 (-1.32)	0.4929 (-0.0221)	0.4435 (-0.0232)

Green: gain Red: decrease Gray: unchanged; all deltas are relative to GMR.

N-acc = TN / negatives. τ is the existence threshold. Localization uses positive ungated candidates.

Full dataset and GEQ Full training mode are distinct. Coverage and temporal limitations: see DATA.md.

Same Standard-trained checkpoints across both Test domains. Training budgets and native seeds differ; see TRAINING.md.